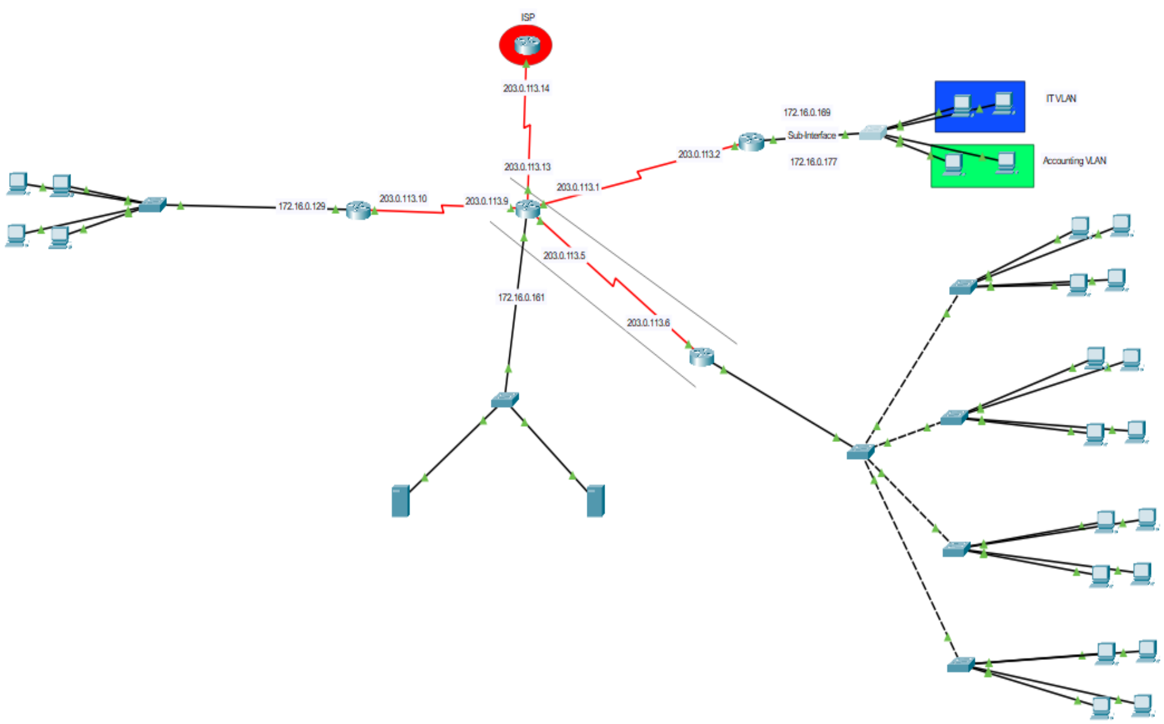
Network-1 Project Documentation Report

Project Number: 5
Project Name : The Distributed E-Commerce Fulfillment Network
Discussion Time : 12:15

Full Topology:

The network is designed as a Wide Area Network (WAN) consisting of four core routers (Fulfillment Center, Shipping Hub, Data Center, Corporate HQ) plus an ISP router. The topology includes switches, end devices (PCs), a DNS server, and a web server. All devices are interconnected via serial and Ethernet links, and static routing is used to achieve full connectivity. Screenshot: Full logical topology from Packet Tracer:



Device & Connection Specifications:

Details of all hardware and physical-layer components used in the topology.

Hardware Inventory

Device Type	Model	Quantity	Notes
Router	2911	3	
Router	ISR4321	2	Used instead of 2911 for implementing VPN
Switch	2950-T	7	
PC / End Host	----------	98	
Server (DNS)	----------	1	
Server (Web)	----------	1	

Cabling & Connection Types

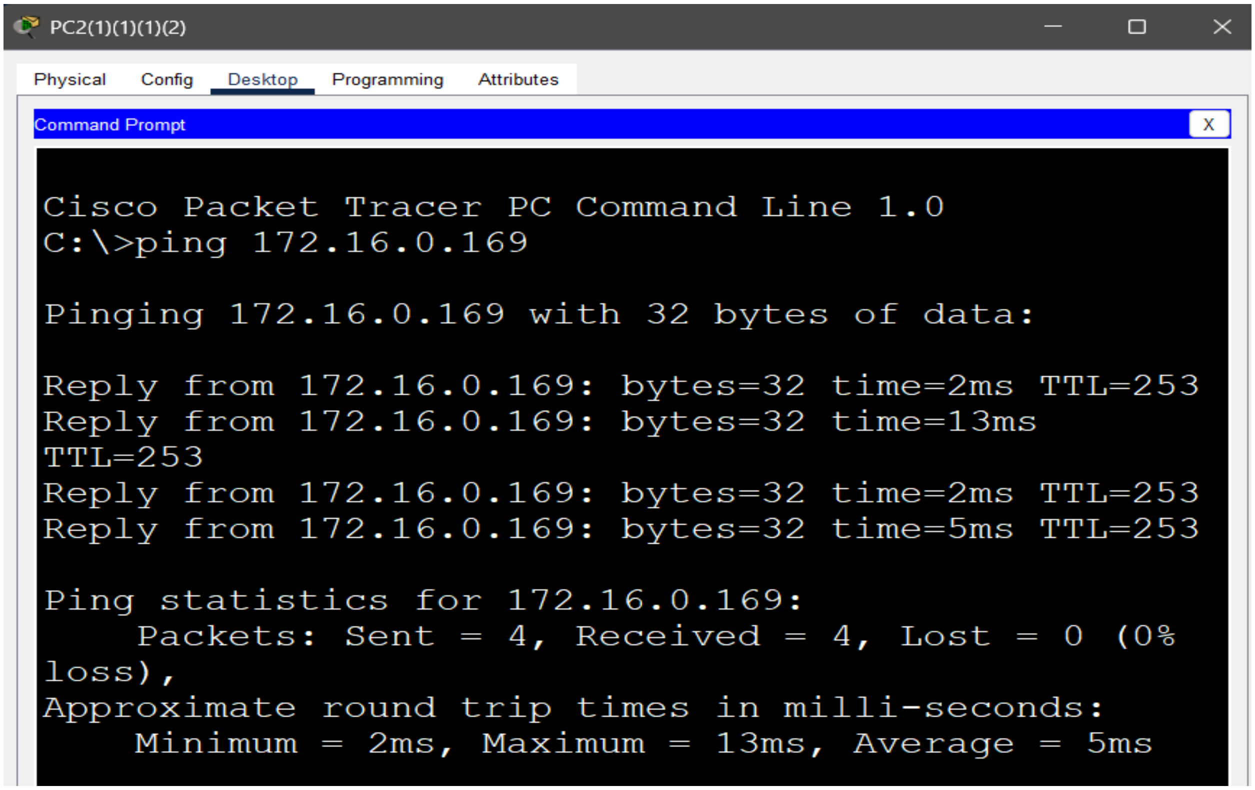
Connection (From → To)	Cable Type
Between Routers	Serial
Router and Switch	Copper Straight-Through
Between Switches	Copper Cross-Over
Switch and PCs(or Servers)	Copper Straight-Through

successful ping between networks:

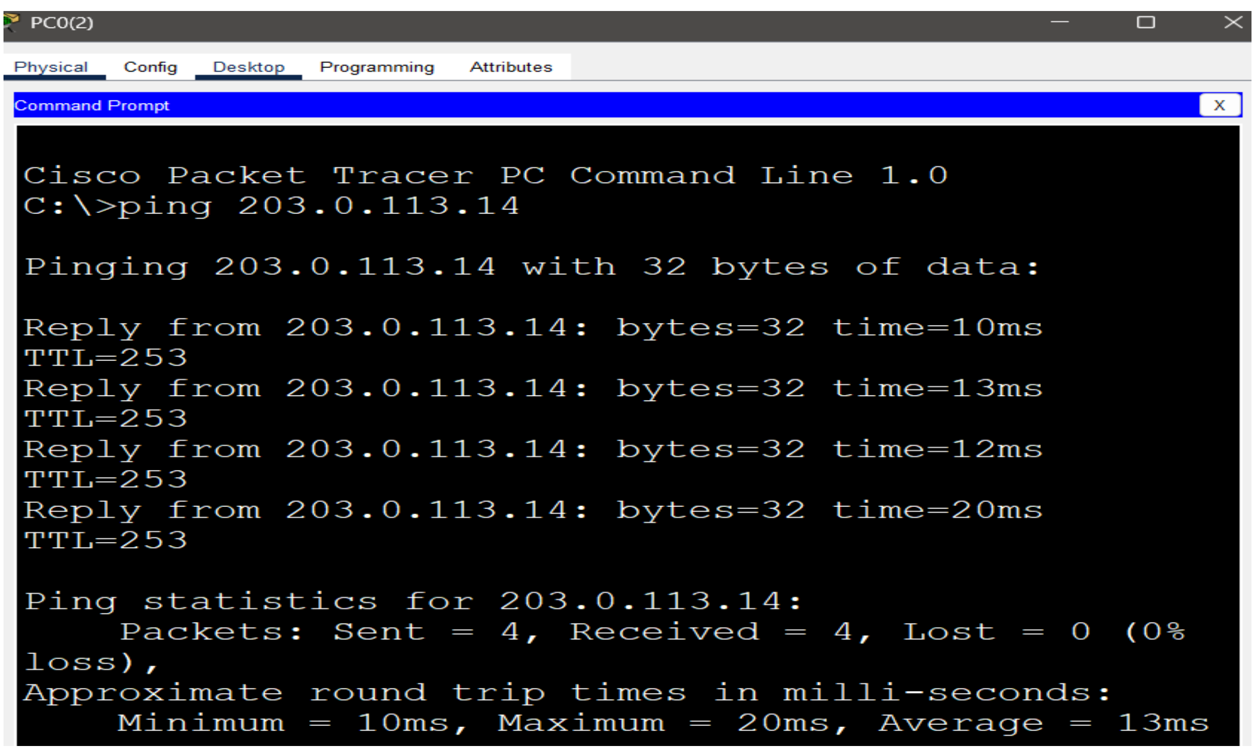
PC (Fulfillment Center) to Web Server (Data Center):

```
C:\>ping 172.16.0.163

Pinging 172.16.0.163 with 32 bytes of data:

Reply from 172.16.0.163: bytes=32 time=2ms
TTL=126
Reply from 172.16.0.163: bytes=32 time=10ms
TTL=126
Reply from 172.16.0.163: bytes=32 time=11ms
TTL=126
Reply from 172.16.0.163: bytes=32 time=1ms
TTL=126

Ping statistics for 172.16.0.163:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 6ms
```

PC (Shipping Hub) to PC (HQ):



Any PC to ISP Router:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0(2)	ISP	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	Web	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1(...	PC0(2)(2)	ICMP		0.000	N	2	(edit)	(delete)

1. VLSM Chart (Internal Networks)

The private IP range 172.16.0.0/16 was divided using VLSM to meet host requirements of each LAN. The following tables summarize the subnet allocation for internal networks and WAN links.

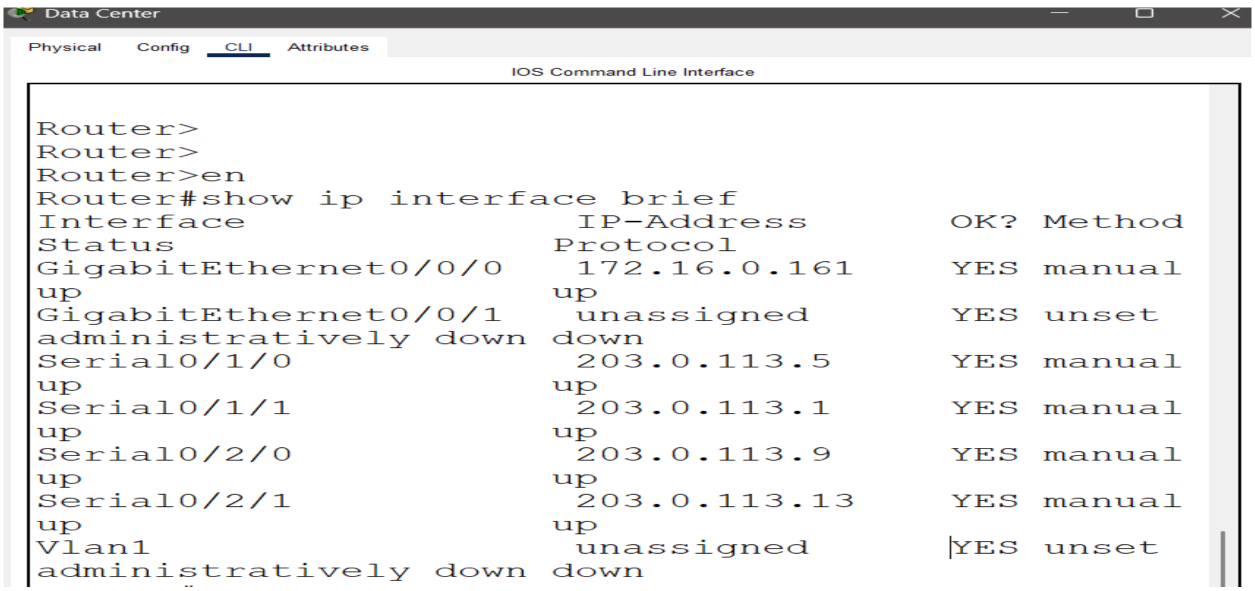
Network Name	**Hosts**	**Subnet**	**Network ID**	**Subnet Mask**	**First useable IP**	**Last useable IP**	**Broadcast**
Fulfillment Center LAN	81	**/25** (126 hosts)	172.16.0.0	255.255.255.128	172.16.0.1	172.16.0.126	172.16.0.127
Shipping Hub LAN	15	**/27** (30 hosts)	172.16.0.128	255.255.255.224	172.16.0.129	172.16.0.158	172.16.0.159
Data Center LAN	3	**/29** (6 hosts)	172.16.0.160	255.255.255.248	172.16.0.161	172.16.0.166	172.16.0.167
HQ VLAN 10	3	**/29** (6 hosts)	172.16.0.168	255.255.255.248	172.16.0.169	172.16.0.174	172.16.0.175
HQ VLAN 20	3	**/29** (6 hosts)	172.16.0.176	255.255.255.248	172.16.0.177	172.16.0.182	172.16.0.183

2. VLSM Chart (WAN Links)

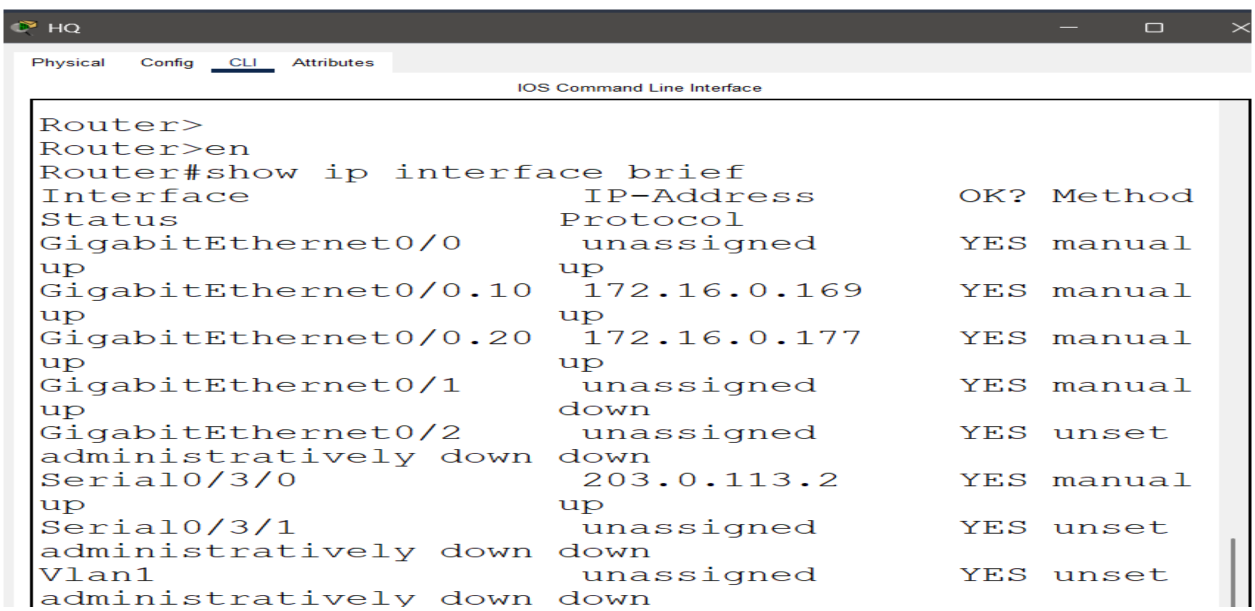
Link	**Network ID**	**Subnet Mask**	**First useable IP**	**Last useable IP**	**Broadcast**
Data Center ↔ Corp HQ	203.0.113.0	255.255.255.252	203.0.113.1	203.0.113.2	203.0.113.3
Data Center ↔ Fulfillment	203.0.113.4	255.255.255.252	203.0.113.5	203.0.113.6	203.0.113.7

Data Center ↔ Shipping Hub	203.0.113.8	255.255.255.252	203.0.113.9	203.0.113.10	203.0.113.11
Data Center ↔ ISP	203.0.113.12	255.255.255.252	203.0.113.13	203.0.113.14	203.0.113.15

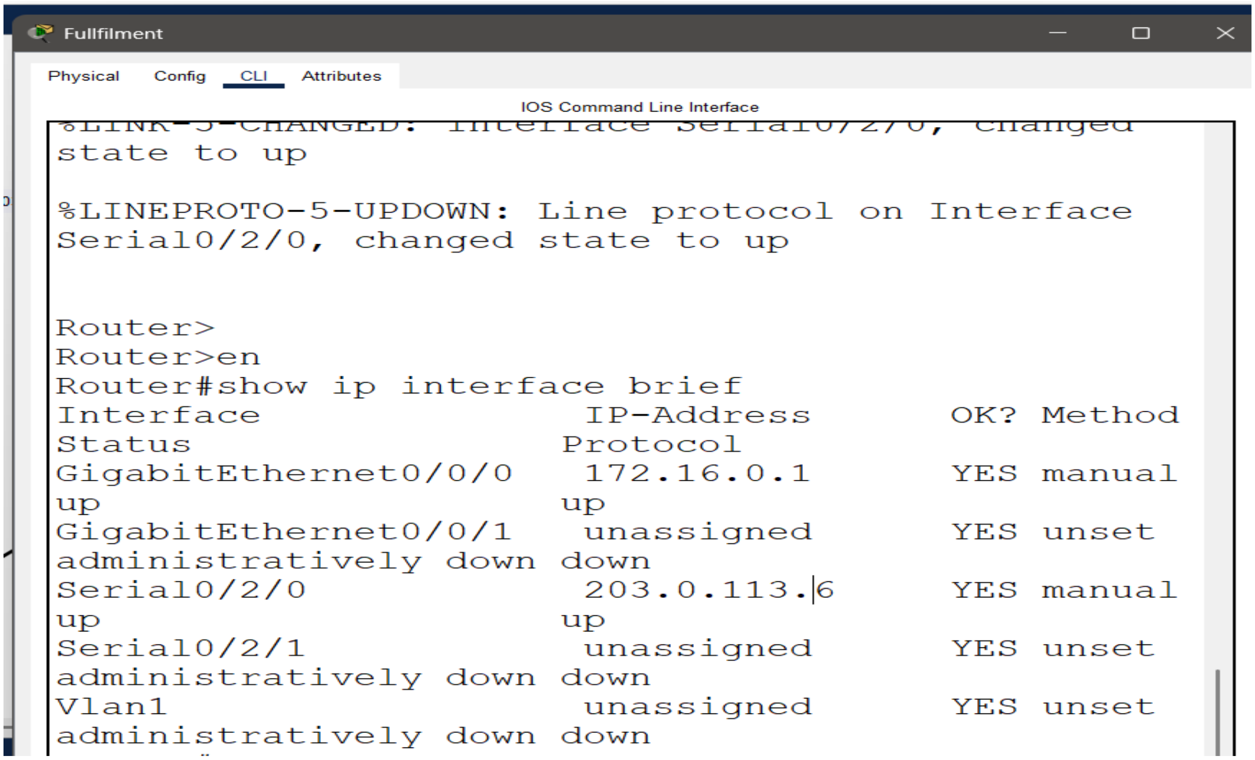
Router IP configurations:

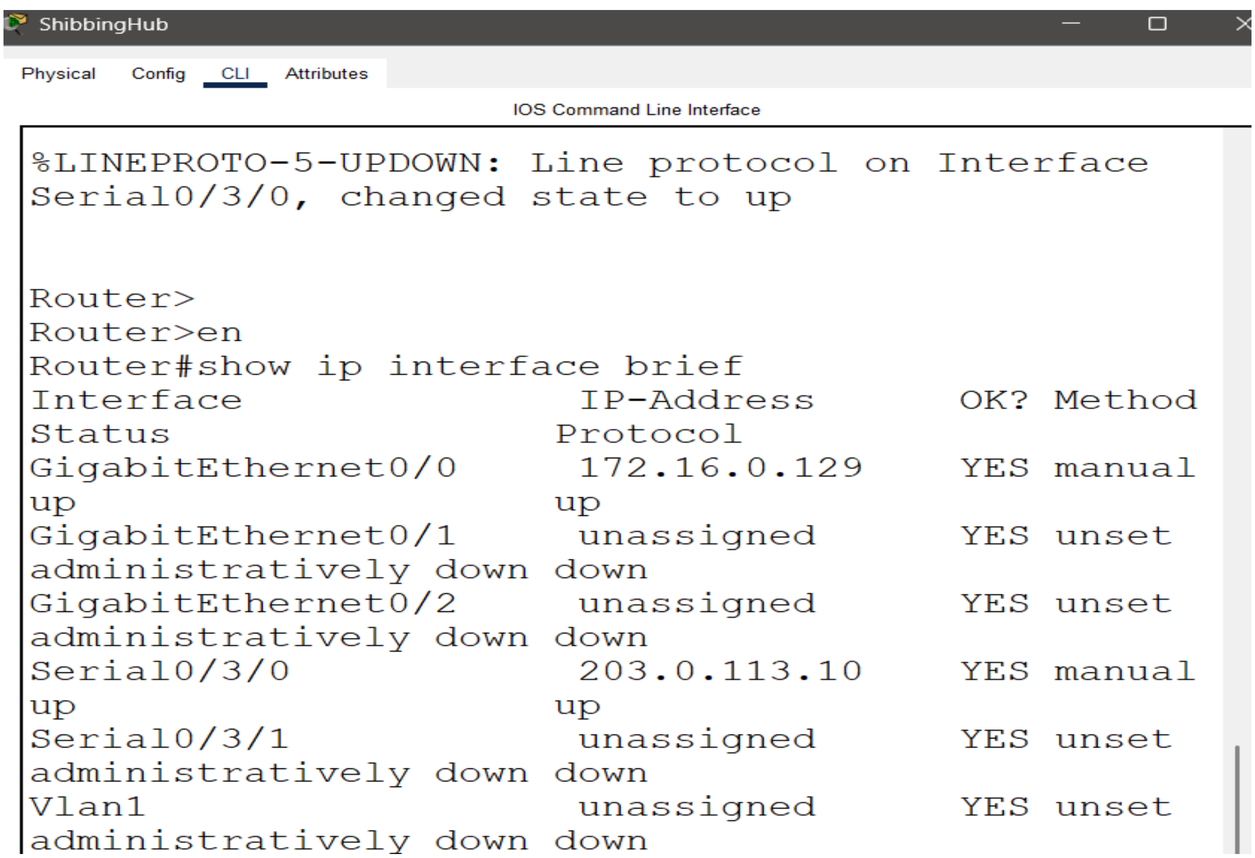


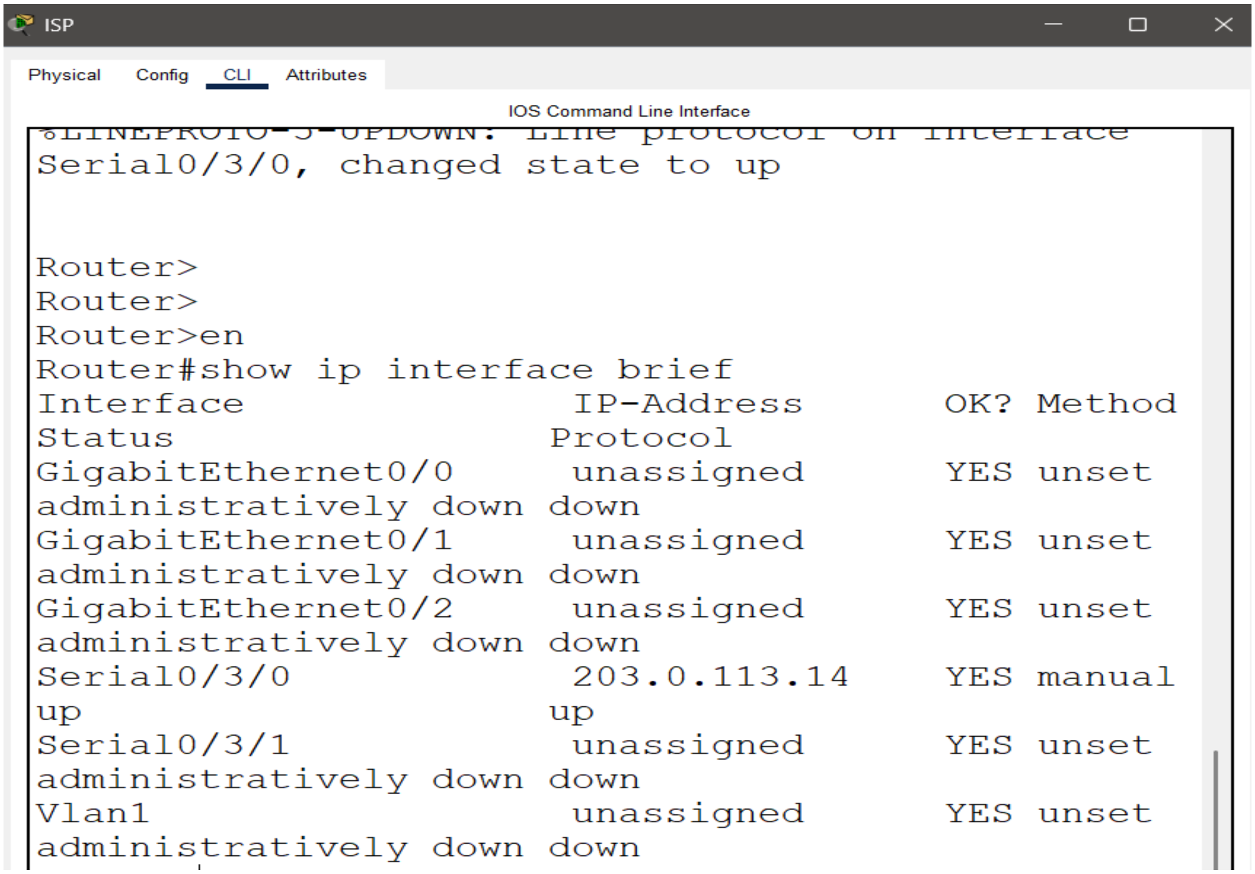
```
Data Center
Physical   Config   CLI   Attributes
IOS Command Line Interface

Router>
Router>
Router>en
Router#show ip interface brief
Interface              IP-Address      OK? Method
Status                 Protocol
GigabitEthernet0/0/0   172.16.0.161    YES manual
up                     up
GigabitEthernet0/0/1   unassigned      YES unset
administratively down  down
Serial0/1/0            203.0.113.5     YES manual
up                     up
Serial0/1/1            203.0.113.1     YES manual
up                     up
Serial0/2/0            203.0.113.9     YES manual
up                     up
Serial0/2/1            203.0.113.13    YES manual
up                     up
Vlan1                  unassigned      YES unset
administratively down  down
```



```
HQ
Physical   Config   CLI   Attributes
IOS Command Line Interface

Router>
Router>en
Router#show ip interface brief
Interface              IP-Address      OK? Method
Status                 Protocol
GigabitEthernet0/0     unassigned      YES manual
up                     up
GigabitEthernet0/0.10  172.16.0.169    YES manual
up                     up
GigabitEthernet0/0.20  172.16.0.177    YES manual
up                     up
GigabitEthernet0/1     unassigned      YES manual
up                     down
GigabitEthernet0/2     unassigned      YES unset
administratively down  down
Serial0/3/0            203.0.113.2     YES manual
up                     up
Serial0/3/1            unassigned      YES unset
administratively down  down
Vlan1                  unassigned      YES unset
administratively down  down
```


Fullfilment
Physical
Config
CLI
Attributes
IOS Command Line Interface
state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up
Router>
Router>en
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 172.16.0.1 YES manual up up
GigabitEthernet0/0/1 unassigned YES unset administratively down down
Serial0/2/0 203.0.113.6 YES manual up up
Serial0/2/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down

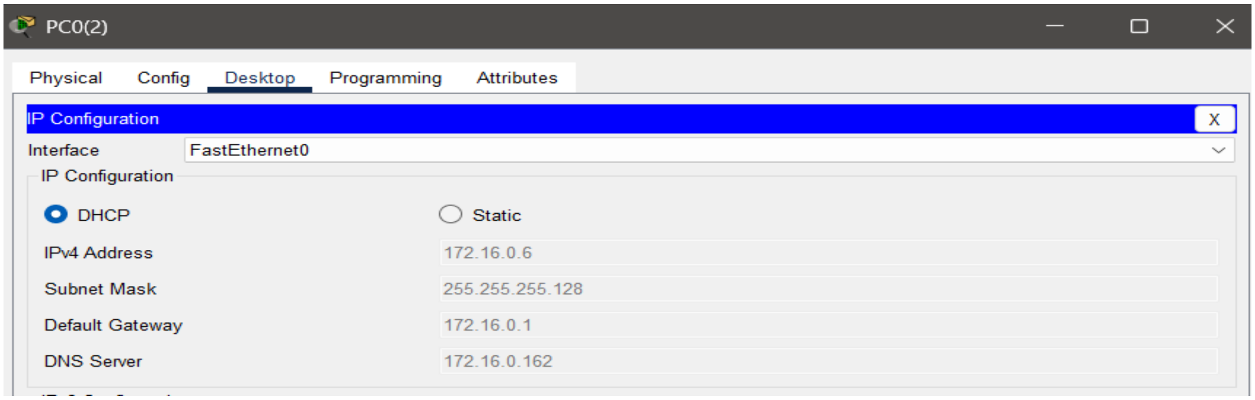

ShibbingHub
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
Router>
Router>en
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.0.129 YES manual up up
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/3/0 203.0.113.10 YES manual up up
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down

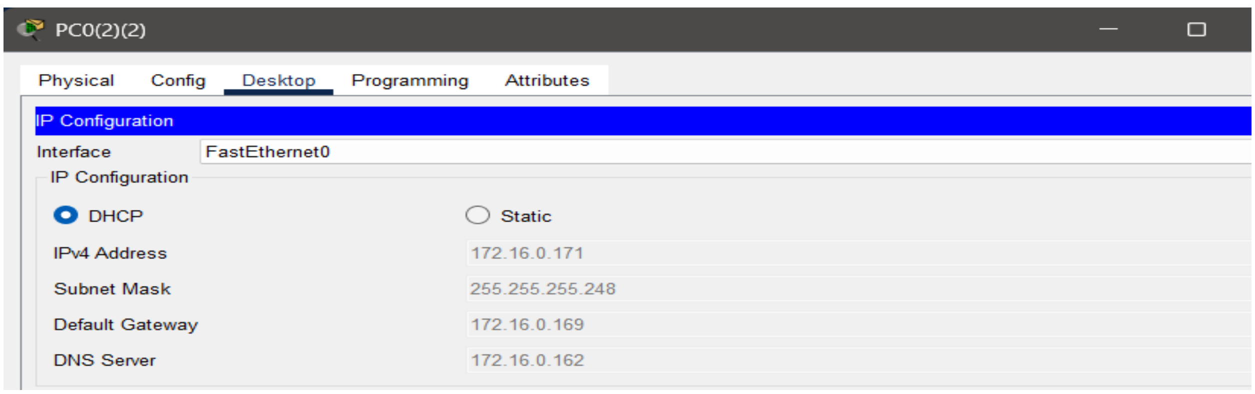


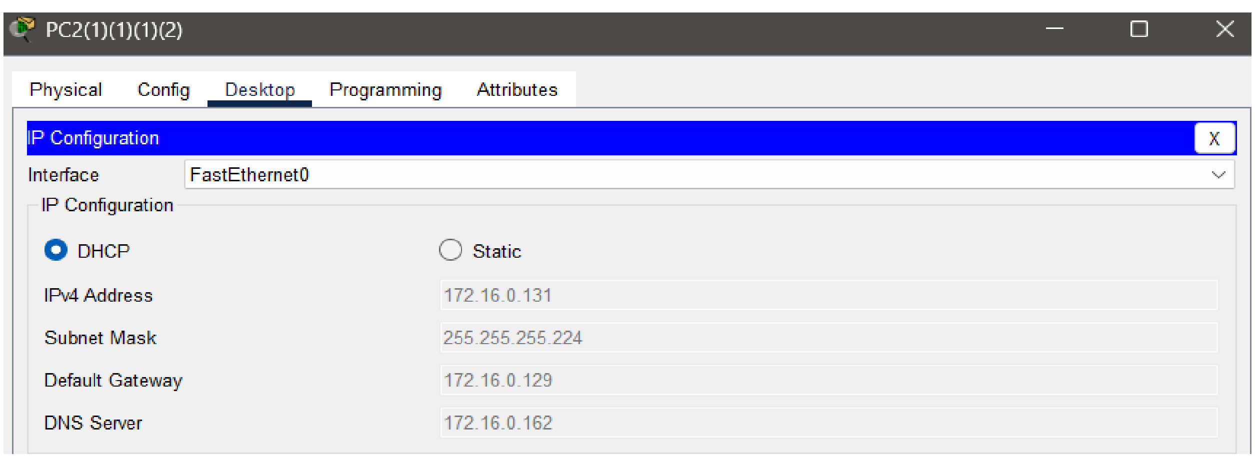
DHCP + sample PC:

DHCP pools were configured on the routers to automatically assign IP addresses to hosts in their respective LANs:

Router	Pool Name	Network	Gateway	DNS Server
Fulfillment	LAN1	172.16.0.0/25	172.16.0.1	172.16.0.162
Shipping Hub	Shipping-Hub	172.16.0.128/27	172.16.0.129	172.16.0.162
Corporate HQ	Vlan1	172.16.0.168/29	172.16.0.169	172.16.0.162
Corporate HQ	Vlan2	172.16.0.176/29	172.16.0.177	172.16.0.162


PC0(2)
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.0.6
Subnet Mask
255.255.255.128
Default Gateway
172.16.0.1
DNS Server
172.16.0.162


PC0(2)(2)
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.0.171
Subnet Mask
255.255.255.248
Default Gateway
172.16.0.169
DNS Server
172.16.0.162


PC2(1)(1)(1)(2)
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.0.131
Subnet Mask
255.255.255.224
Default Gateway
172.16.0.129
DNS Server
172.16.0.162

ShibbingHub

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface
Serial0/3/0, changed state to up

Router#
Router#
Router#show ip dhcp pool

Pool Shipping-Hub :
 Utilization mark (high/low)      : 100 / 0
 Subnet size (first/next)         : 0 / 0
 Total addresses                  : 30
 Leased addresses                 : 3
 Excluded addresses               : 0
 Pending event                    : none

 1 subnet is currently in the pool
 Current index         IP address range
Leased/Excluded/Total
 172.16.0.129          172.16.0.129      -
172.16.0.158      3     / 0       / 30
```

Fullfilment

Physical Config CLI Attributes

IOS Command Line Interface

```
Serial0/2/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
Serial0/2/0, changed state to up

Router#
Router#
Router#show ip dhcp pool

Pool lan1 :
 Utilization mark (high/low)      : 100 / 0
 Subnet size (first/next)         : 0 / 0
 Total addresses                  : 126
 Leased addresses                 : 3
 Excluded addresses               : 0
 Pending event                    : none

 1 subnet is currently in the pool
 Current index         IP address range
Leased/Excluded/Total
 172.16.0.1            172.16.0.1        -
172.16.0.126      3     / 0       / 126
```

HQ

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip dhcp pool

Pool vlan1 :
 Utilization mark (high/low)      : 100 / 0
 Subnet size (first/next)         : 0 / 0
 Total addresses                  : 6
 Leased addresses                 : 2
 Excluded addresses               : 0
 Pending event                    : none

 1 subnet is currently in the pool
 Current index          IP address range
Leased/Excluded/Total
 172.16.0.169           172.16.0.169     -
172.16.0.174      2     / 0     / 6

Pool vlan2 :
 Utilization mark (high/low)      : 100 / 0
 Subnet size (first/next)         : 0 / 0
 Total addresses                  : 6
 Leased addresses                 : 1
 Excluded addresses               : 0
 Pending event                    : none
```

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VLAN table:

On the HQ switch, two VLANs were created: VLAN 10 (IT) and VLAN 20 (Accounting). Access ports were assigned accordingly:

VLAN ID	VLAN Name	State	Ports
10	IT	Active	f0/1, f0/2
20	Accounting	Active	f0/3, f0/4

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- ---------
-------------------------------
1    default                          active    Fa0/6, Fa0/7,
Fa0/8, Fa0/9
                                                Fa0/10,
Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14,
Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18,
Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22,
Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   it                               active    Fa0/1, Fa0/2
20   accounting                       active    Fa0/3, Fa0/4
```

NAT translation table:

NAT was implemented on the Data Center router

Detail	Value
Router performing NAT	Data Center
NAT Type (Static / Dynamic / PAT)	Static for Web Server / PAT for End devices
Inside Interface(s)	Gig0/0/0 (Static) / Se0/1/0, Se0/1/1, Se0/2/0
Outside Interface(s)	Se0/2/1
NAT Pool / Overload Address	203.0.113.13(Overload)

Data Center

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside
global
icmp 203.0.113.13:1     172.16.0.5:1       203.0.113.14:1
203.0.113.14:1
icmp 203.0.113.13:2     172.16.0.5:2       203.0.113.14:2
203.0.113.14:2
icmp 203.0.113.13:3     172.16.0.5:3       203.0.113.14:3
203.0.113.14:3
icmp 203.0.113.13:4     172.16.0.5:4       203.0.113.14:4
203.0.113.14:4
---  203.0.113.20       172.16.0.163       ---                ---
```

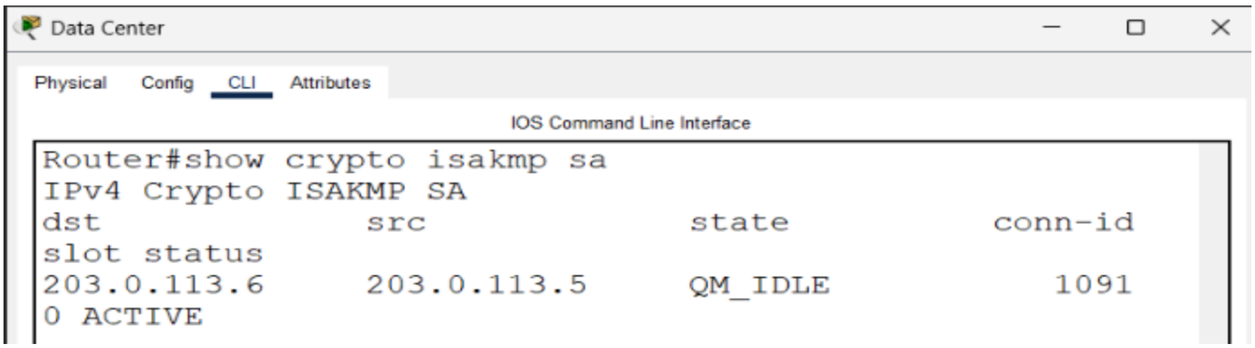
```
Router#show ip nat statistics
NAT: s=172.16.0.4->203.0.113.13, d=203.0.113.14 [29]

NAT*: s=203.0.113.14, d=203.0.113.13->172.16.0.4 [44]

NAT: s=172.16.0.4->203.0.113.13, d=203.0.113.14 [30]

NAT*: s=203.0.113.14, d=203.0.113.13->172.16.0.4 [45]

NAT: s=172.16.0.4->203.0.113.13, d=203.0.113.14 [31]

NAT*: s=203.0.113.14, d=203.0.113.13->172.16.0.4 [46]

NAT: s=172.16.0.4->203.0.113.13, d=203.0.113.14 [32]

NAT*: s=203.0.113.14, d=203.0.113.13->172.16.0.4 [47]

NAT: expiring 203.0.113.13 (172.16.0.4) icmp 26 (26)

NAT: expiring 203.0.113.13 (172.16.0.4) icmp 27 (27)

NAT: expiring 203.0.113.13 (172.16.0.4) icmp 28 (28)

NAT: expiring 203.0.113.13 (172.16.0.4) icmp 29 (29)
```

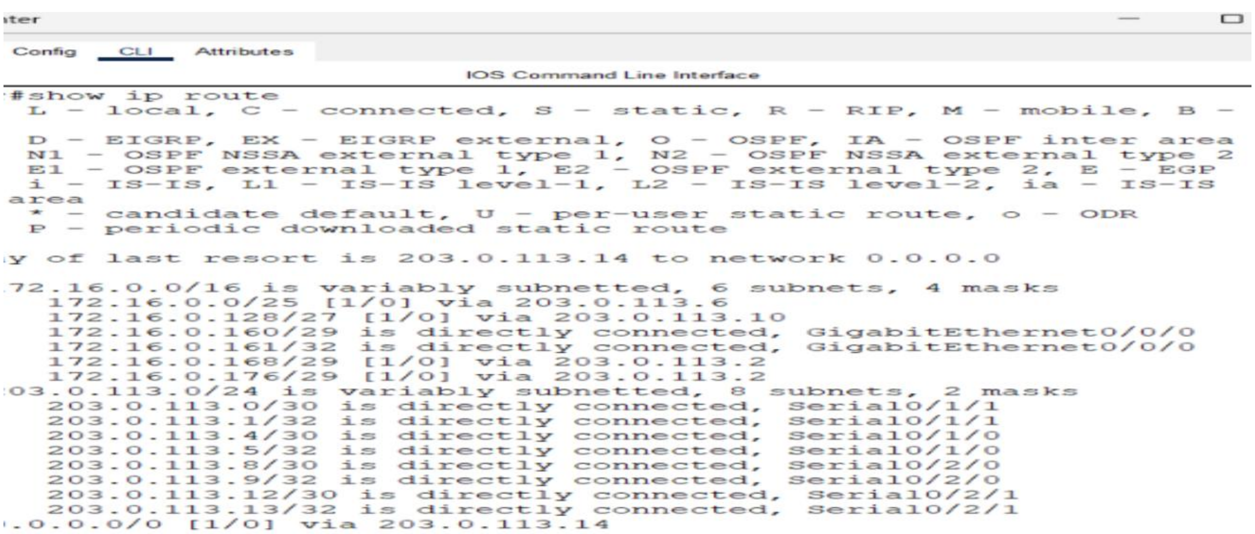
VPN status + successful ping:

A site-to-site IPsec VPN tunnel was established between the Fulfillment Center router and the Data Center router.

VPN (Site-to-Site)

Detail	Value
VPN Tunnel Start (Router)	Fullfilment Center
VPN Tunnel End (Router)	Data Center
Interesting Traffic (ACLs)	172.16.0.0/25 and 172.16.0.160 /29
Pre-shared Key	helwan





Data Center

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show crypto ipsec sa

interface: Serial0/1/0
    Crypto map tag: MAP, local addr 203.0.113.5

   protected vrf: (none)
   local  ident (addr/mask/prot/port):
(172.16.0.160/255.255.255.248/0/0)
   remote  ident (addr/mask/prot/port):
(172.16.0.0/255.255.255.128/0/0)
   current_peer 203.0.113.6 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 8, #pkts encrypt: 8, #pkts digest: 0
   #pkts decaps: 12, #pkts decrypt: 12, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0
```

Routing table from each router:

Static routes were configured on every router to ensure reachability between all networks. Default routes were used on the branch routers pointing to the Data Center core router.

Data Center

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.14 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 6 subnets, 4 masks
S       172.16.0.0/25 [1/0] via 203.0.113.6
S       172.16.0.128/27 [1/0] via 203.0.113.10
C       172.16.0.160/29 is directly connected, GigabitEthernet0/0/0
L       172.16.0.161/32 is directly connected, GigabitEthernet0/0/0
S       172.16.0.168/29 [1/0] via 203.0.113.2
S       172.16.0.176/29 [1/0] via 203.0.113.2
     203.0.113.0/24 is variably subnetted, 8 subnets, 2 masks
C       203.0.113.0/30 is directly connected, Serial0/1/1
L       203.0.113.1/32 is directly connected, Serial0/1/1
C       203.0.113.4/30 is directly connected, Serial0/1/0
L       203.0.113.5/32 is directly connected, Serial0/1/0
C       203.0.113.8/30 is directly connected, Serial0/2/0
L       203.0.113.9/32 is directly connected, Serial0/2/0
C       203.0.113.12/30 is directly connected, Serial0/2/1
L       203.0.113.13/32 is directly connected, Serial0/2/1
S*   0.0.0.0/0 [1/0] via 203.0.113.14
```

Fullfilment

Physical Config CLI Attributes

IOS Command Line Interface

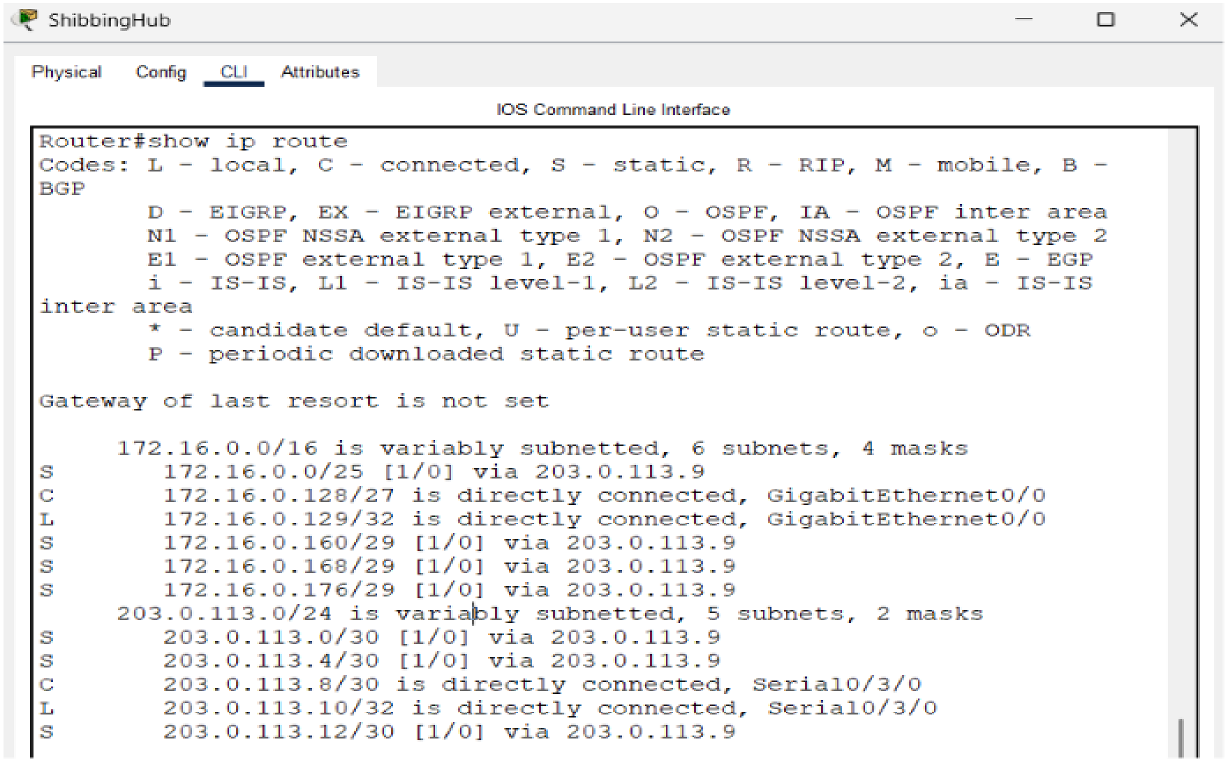
```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.5 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 6 subnets, 4 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0/0
S       172.16.0.128/27 [1/0] via 203.0.113.5
S       172.16.0.160/29 [1/0] via 203.0.113.5
S       172.16.0.168/29 [1/0] via 203.0.113.5
S       172.16.0.176/29 [1/0] via 203.0.113.5
     203.0.113.0/24 is variably subnetted, 5 subnets, 2 masks
S       203.0.113.0/30 [1/0] via 203.0.113.5
C       203.0.113.4/30 is directly connected, Serial0/2/0
L       203.0.113.6/32 is directly connected, Serial0/2/0
S       203.0.113.8/30 [1/0] via 203.0.113.5
S       203.0.113.12/30 [1/0] via 203.0.113.5
S*   0.0.0.0/0 [1/0] via 203.0.113.5
```

HQ

Physical Config CLI Attributes

IOS Command Line Interface

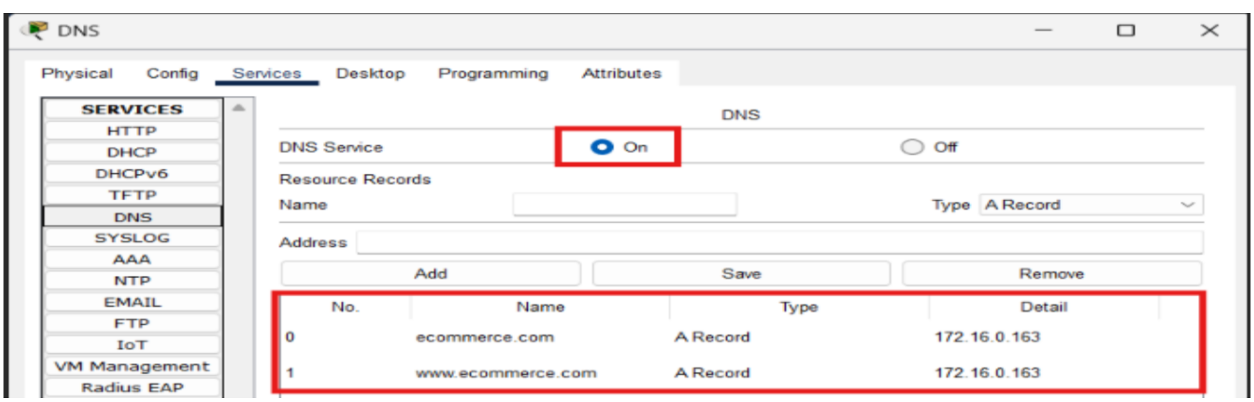
```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 7 subnets, 4 masks
S       172.16.0.0/25 [1/0] via 203.0.113.1
S       172.16.0.128/27 [1/0] via 203.0.113.1
S       172.16.0.160/29 [1/0] via 203.0.113.1
C       172.16.0.168/29 is directly connected, GigabitEthernet0/0.10
L       172.16.0.169/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.176/29 is directly connected, GigabitEthernet0/0.20
L       172.16.0.177/32 is directly connected, GigabitEthernet0/0.20
     203.0.113.0/24 is variably subnetted, 5 subnets, 2 masks
C       203.0.113.0/30 is directly connected, Serial0/3/0
L       203.0.113.2/32 is directly connected, Serial0/3/0
S       203.0.113.4/30 [1/0] via 203.0.113.1
S       203.0.113.8/30 [1/0] via 203.0.113.1
S       203.0.113.12/30 [1/0] via 203.0.113.1
```

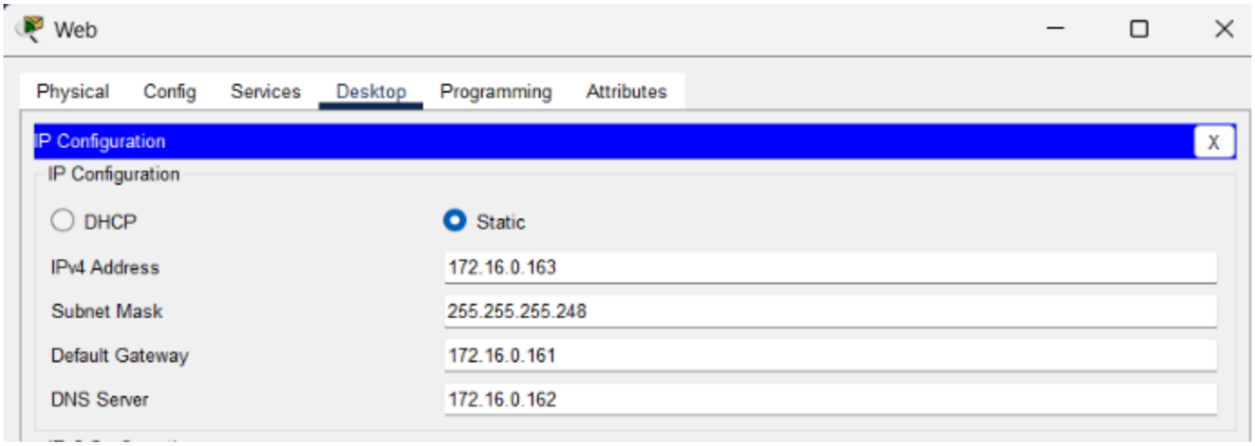


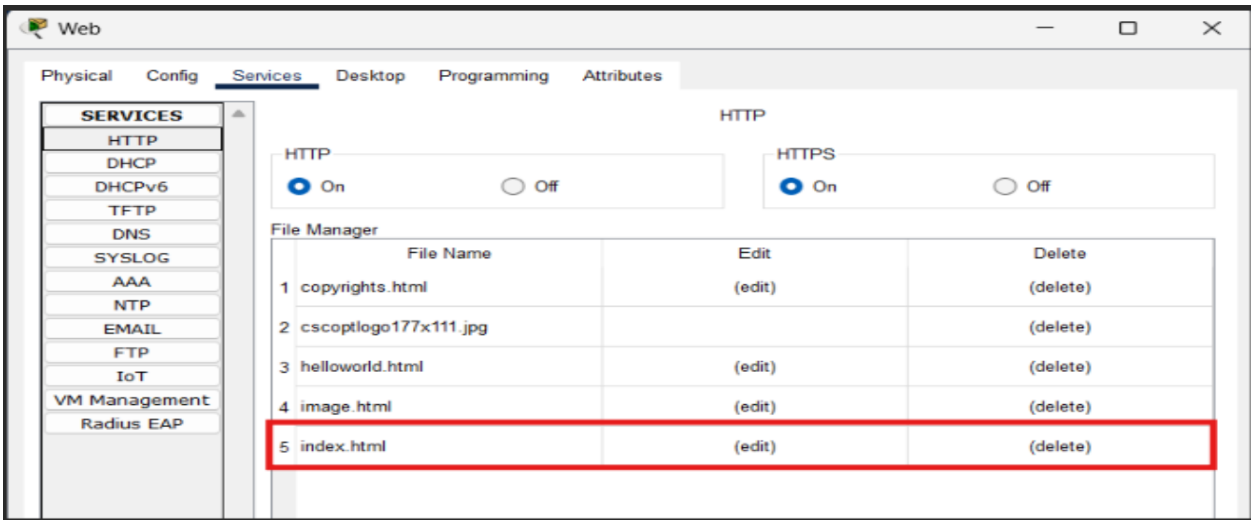
DNS & WEB Servers :

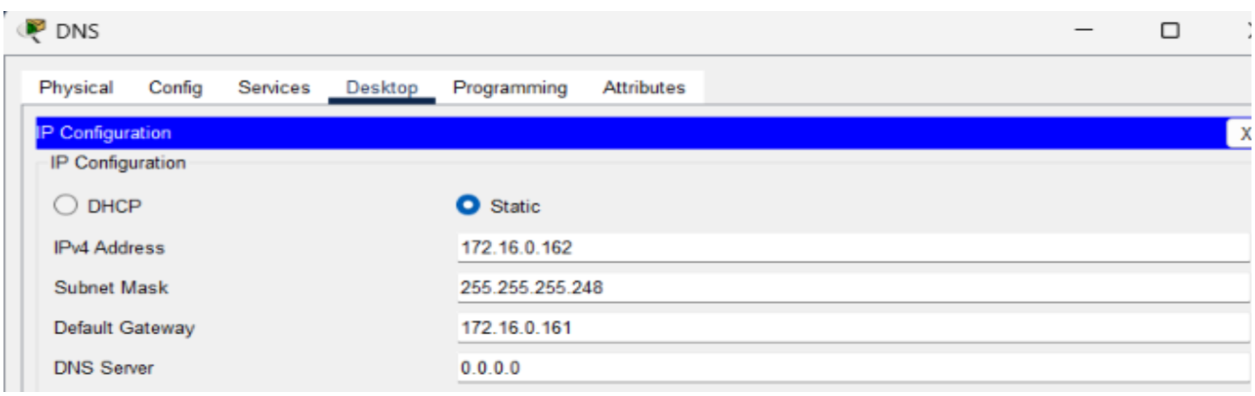
DNS Server

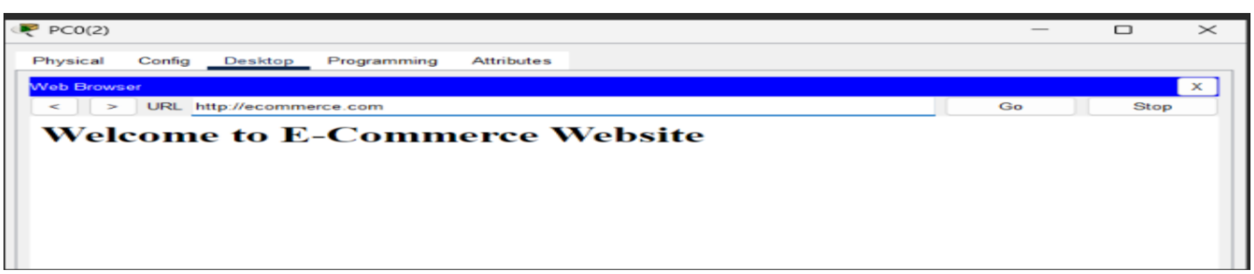
Detail	Value
DNS Server IP Address	172.16.0.162
Domain Name configured	ecommerce.com
Points to (Web Server IP)	172.16.0.163




Web
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
172.16.0.163
Subnet Mask
255.255.255.248
Default Gateway
172.16.0.161
DNS Server
172.16.0.162


Web
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
HTTP
HTTP
On
Off
HTTPS
On
Off
File Manager
File Name
Edit
Delete
1 copyrights.html (edit) (delete)
2 cscoptlogo177x111.jpg (delete)
3 helloworld.html (edit) (delete)
4 image.html (edit) (delete)
5 index.html (edit) (delete)


DNS
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
172.16.0.162
Subnet Mask
255.255.255.248
Default Gateway
172.16.0.161
DNS Server
0.0.0.0

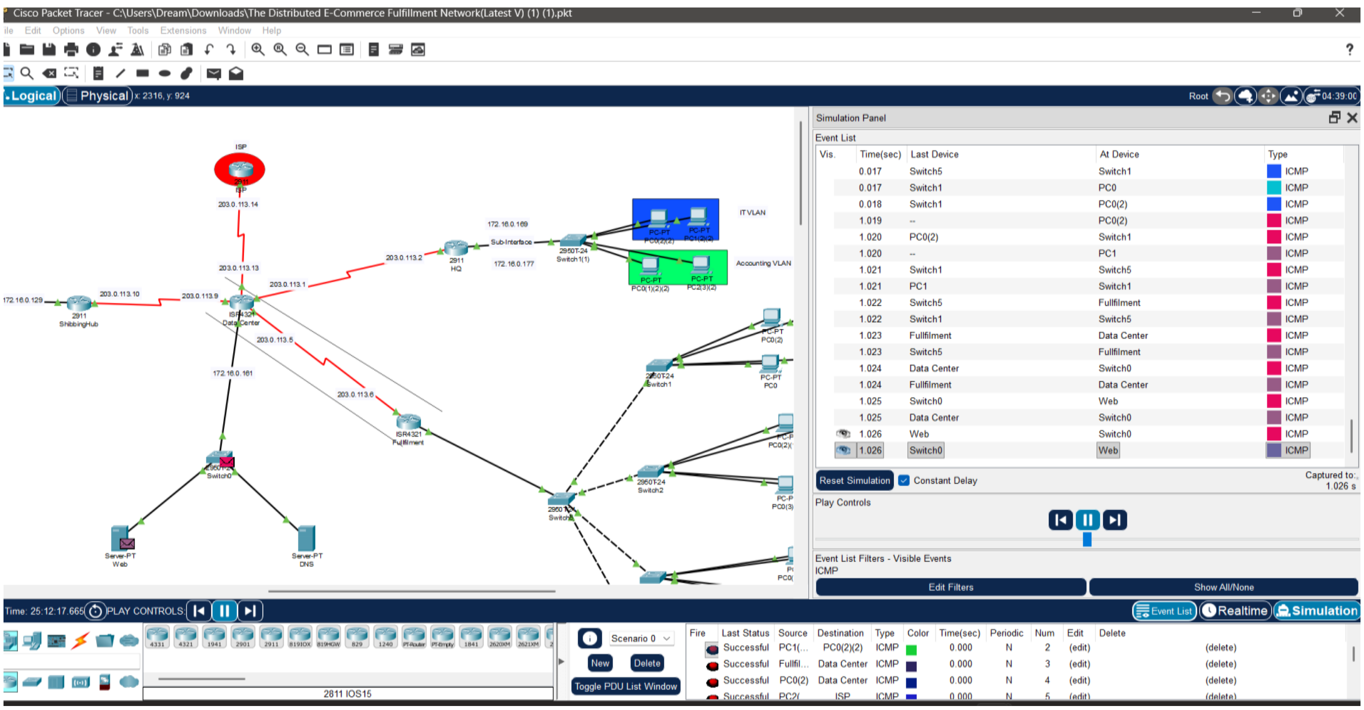

PC0(2)
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://ecommerce.com
Go
Stop
Welcome to E-Commerce Website

ICMP Testing:

Connectivity between all networks was verified using ICMP (ping). A successful ping from PC1 (Fulfillment Center, 172.16.0.6) to Web Server (172.16.0.163) confirms end-to-end reachability across the WAN.

Successful ping: Reply from 172.16.0.163 with 0% loss

Simulation mode: Packet path traced through routers, confirming static routes and NAT are working correctly.



Thank you, Eng. Ibrahim, for reading all the Documentation to the end ☺.